

Pintu Singh

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PROFILE

Sr.Data Engineer skilled in building and optimizing large-scale data pipelines, ETL processes, and data warehouse solutions. Skilled in **Python, SQL, Spark, Azure, Databricks, and Kafka**. Developed multi-terabyte scalable big data solutions for and Komodo health and UnitedHealth Group (UHG), a Fortune 5 company.

TECHNICAL SKILLS

Programming Languages: Python, C++
Big Data Technologies: Spark, PySpark, Spark SQL, YARN, Kubernetes, Hadoop, Hive, Impala
Streaming Technology: Kafka, Spark structured streaming
Cloud computing: ADF, Databricks, ADLS, Azure Synapse, Amazon S3, AWS Glue, AWS EMR
Data Engineering Tools: Airflow, Azure DevOps, Data Modelling, ETL/ELT data Pipeline
AI/Agentic AI: LLMs, Generative AI, RAG, Prompt Engineering, LangChain, LangGraph, OpenAI API, MCP
AI Deployment/Infra: Docker, Kubernetes, FastAPI, Vector Databases, Hugging Face, vLLM
DataBases: Snowflake, Teradata SQL, MySQL, Spark SQL

WORK EXPERIENCE

Komodo Health

Noida, India

Data Engineer 3

06/2025 – present

- Reduced 90% of on-call interventions and improved MTTR by 80% by building an AI-powered incident diagnosis engine using RAG, LLM tool-calling, and historical pipeline intelligence to automate root-cause analysis and generate DDL remediation scripts for data migration failures.
- Developed, tested, and maintained RESTful APIs within a healthcare data platform, contributing to scalable and secure data integration across clinical and provider systems
- Diagnosed and resolved critical API issues across staging and production environments, reducing system downtime and improving platform stability by 20%.

Optum (UnitedHealth Group)

Noida, India

Senior Data Engineer

07/2024 - 03/2025

- Optimized overall process performance through **Spark performance tuning**, improving job run times by 20% and efficiently managing a **12TB** dataset containing approximately **10 billion** records.
- Implemented Spark optimization techniques such as **caching, multithreading, and broadcast** joins, resulting in a 20% decrease in processing time for handling a daily load of around **2 million** records.
- Successfully migrated legacy on-premise processes to the cloud using Spark, resulting in a **20% reduction in processing time**.
- Developed multiple automated data pipelines to fetch data from various **Kafka** topics.
- Optimized queries and data processing**, resulting in 30% faster data retrieval and analysis, supporting timely and accurate reporting.

Optum (UnitedHealth Group)

Noida, India

Data Engineer

09/2021 - 01/2024

- Created ETL workflows to extract data from 20+ sources**, transforming it into a standardized format and loading it into a data warehouse, improving data integration and accessibility by 25%.
- Implemented **data quality checks** in Apache Airflow DAGs, ensuring 99.6% accuracy in data transformations and load operations.
- Automated incident logging for ETL pipeline failures using machine learning**, reducing manual intervention by 60% and improving response times.
- Analyzed data to solve a **wide variety of business problems**, creating data visualizations that drove strategic direction and improved decision-making processes.
- Demonstrated reliability and expertise through multiple **on-call** rotations, effectively resolving critical production issues to ensure uninterrupted system functionality.

EDUCATION

NIT Silchar

CGPA: 7.36

B.Tech in Computer Science and Engineering

07/2017 - 05/2021

PUBLICATIONS

- Multimodal Recipe Recommendation System Using Deep Learning and Rule-Based Approach (Show publication)
- Abstractive Text Summarization Approaches with Analysis of Evaluation Techniques (Show publication)